Mid-Semester Question Bank

Units I–IV and Unit V (part) • Lecture packages L1–L10
CSAI2017P: Applied Machine Learning

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How to use this

This is not homework. Nothing in here is marked, nothing is collected, and you may work through it in any order. It exists because you deserve to know what the mid-semester syllabus means in practice rather than in syllabus headings.

Every question below is one that could reasonably be set on your paper. They are drawn from the seven lecture packages you have been taught — the worked examples, the key ideas, the cautions and the practice questions — with the numbers changed so that you have to do the work rather than recall the answer.

Three tags, and what they mean

[⋆⋆⋆] **Near certain.** This was a board item, a predict-first reveal, or a named examinable skill. If it does not appear on the paper in some form, I will be mildly surprised.
[⋆⋆] **Likely.** A worked example or a "key idea" box from the notes.
[⋆] **Possible.** Taught, examinable, but peripheral.
[**NUM**] **Numerical.** Compute it by hand, on paper, showing every step. These are the questions that separate scripts, and they are the ones students skip in revision because reading feels like progress and arithmetic feels like work. Do them.

Do the numericals with a pen

There are **63** [**NUM**] questions here. Reading a worked solution produces the feeling of understanding and almost none of the substance. You will find out in the examination hall which of the two you have. Work each one on paper, *then* check it against the answers.
The answers are in `A1-Bank-Answers.pdf`, in this folder — all 314 of them, with full working for every numerical. Open them *after* you have worked the question, not instead of working it.

The paper you are preparing for

Section	Questions	Rule	Marks
A	5 short answers	all compulsory	$5 \times 2 = 10$
B	4 medium questions	attempt any 3	$3 \times 5 = 15$
C	1 long question	compulsory	$1 \times 10 = 10$
Total raw			**35**
Scaled to the mid-semester weight			**20**
Duration			**90 min**

Neither the syllabus nor the pattern is confirmed yet — read this

The examination cell has not yet issued the mid-semester notice. Both the paper pattern above *and* the exact unit coverage are **expectations, not announcements**, and I will tell you the moment either is confirmed.

Why Units I–IV plus Unit V to L10 is the working assumption. A mid-semester paper can only examine what has been taught by the time you sit it. The calendar delivers lecture packages **L1–L10** before the mid-semester point — Units I to IV in full, and Unit V as far as logistic regression. That is where the paper is expected to come from, and it is what this bank covers. **Unit V stops at L10:** multiple linear regression (L11) and regularization (L12) are not here and are not expected.

What to do if it turns out otherwise. This bank is organised strictly by unit, so it survives either kind of surprise:

- **Narrower** (say it stops at Unit IV, or at Unit III): work the parts that are in and skip the rest. Nothing else changes; the questions you skip become end-semester revision, which you will need anyway.
- **Wider** (reaching past L10 into L11 or L12): everything here still applies — nothing in Units I–V(pt) stops being examinable — and I will issue a supplement rather than reissuing this document.
- **A different section pattern**: only the two mock papers are affected, and only in their shape. The questions inside them stay valid, because a 5-mark question does not become wrong when it is worth 6.

None of these is a reason to delay starting. The overlap between any plausible mid-semester syllabus and this bank is large, and every question in it is end-semester material too.

Coverage map

Unit	What it covers	Lect.	Questions here	Weight to expect
I	ML types, the Python stack, generalisation, overfitting, leakage	L1	24 (U1.1–U1.24)	light — at most 1 short, or a part of one
II	MSE, MAE, RMSE, Huber, cross-entropy, hinge, triplet	L2	38 (U2.1–U2.38)	light–moderate — expect 1 short, possibly 1 medium
III	GD, SGD, mini-batch, momentum, Adagrad, RMSprop, Adadelta, Adam	L3, L4	52 (U3.1–U3.52)	moderate — expect 1 short and up to 1 medium
IV	Cleaning, outliers, transformation, features, encoding, PCA, selection, CV, imbalance	L5, L6, L7	96 (U4.1–U4.96)	**heavy** — expect 1–2 short and 1 medium, and it is a likely source for the long question
V (pt)	Least squares, MLE, R^2, adequacy, over-fitting, logistic regression	L8, L9, L10	84 (U5.1–U5.84)	**heavy** — expect 1 short and 1–2 medium, and it is the other likely source for the long question
—	Two full mock papers (B and C)	—	20	—

The “weight to expect” column is *indicative*, not a prediction of the actual paper — the entries deliberately overlap, because which unit supplies which question is not knowable in advance. What it does say is reliable: weight follows **teaching time**, not unit numbers. Units IV and V are three lecture packages each and will dominate; Units I and II are one package each and will not carry much. Budget your revision that way. If the coverage is announced differently, this column is the only part of the map that changes.

Unit I — Introduction and the Python ML stack (L1)

§A — Two-mark questions

U1.1 [⋆⋆⋆] State Mitchell's definition of machine learning, naming T, P and E. Then fill in all three for a system that predicts whether a student will pass a course.

U1.2 [⋆⋆⋆] Classify each as supervised, unsupervised or reinforcement learning, and name the training signal: (a) forecasting monthly rainfall from 50 years of records; (b) grouping 20,000 news articles by topic with no labels; (c) training an agent to play Pong from the score alone.

U1.3 [⋆⋆] State the containment relationship between artificial intelligence, machine learning and deep learning. Give one task where deep learning is *not* the right choice, with a reason.

U1.4 [⋆⋆] Give three situations in which you should *not* use machine learning, one line each.

U1.5 [⋆⋆⋆] Name the job of NumPy, Pandas and scikit-learn in one line each.

U1.6 [⋆⋆⋆] Explain what `fit`, `transform` and `predict` each do. State why `fit_transform` must never be called on test data.

U1.7 [⋆⋆] What are the "first four calls, every time" on a new DataFrame, and what does each tell you?

U1.8 [⋆] State the NumPy axis mnemonic. For `a` of shape $(3, 4)$, give the shape of `a.sum(axis=0)` and `a.sum(axis=1)`.

U1.9 [⋆⋆] Define overfitting in terms of two numbers. State the single diagnostic that reveals it.

U1.10 [⋆⋆⋆] Define data leakage in one sentence. Give two examples.

U1.11 [⋆⋆] Why is a three-way train/validation/test split needed once you start comparing models? Answer in three lines.

U1.12 [⋆] A classmate reports "accuracy 95.33%" from a single 25% test split of 150 rows. State what is wrong with the way that number is *written*, and what should replace it.

U1.13 [⋆⋆] Name three model families that need feature scaling and two that do not, with the reason each way.

§B — Five-mark questions

U1.14 [⋆⋆⋆][**NUM**] Five training flowers are recorded as (petal length, petal width) in cm with their species:

(1.5, 0.3) setosa	(1.4, 0.2) setosa	(4.8, 1.5) versicolor
(4.2, 1.2) versicolor	(5.9, 2.4) virginica	

A new flower measures (4.6, 1.4).

(a) Compute the Euclidean distance to each of the five, to four decimals.

(b) Give the 1-NN prediction.

(c) Give the 3-NN prediction and the vote.

(d) State what would change if petal width had been recorded in millimetres instead of centimetres, and why.

U1.15 [⋆ ⋆ ⋆][**NUM**] A model predicts $\hat{y} = (4.5, 2.5, 7.0, 5.0)$ against true values $y = (5, 2, 8, 4)$.

(a) Compute MSE, MAE and RMSE.

(b) One prediction is now changed: $\hat{y}_3$ becomes 1.0. Recompute all three.

(c) State the factor by which each metric grew, and explain the ordering.

U1.16 [⋆⋆][**NUM**] A dataset has 569 rows. State (a) the size of a 20% test split; (b) the training and validation fold sizes under 10-fold cross-validation; (c) how many model fits 10-fold CV costs; (d) the number of row-fits, using $n(k-1)$.

U1.17 [⋆ ⋆ ⋆] The table below was measured on 30 points from a gentle curve with 40% held out.

Polynomial degree	**Train RMSE**	**Test RMSE**
1	1.32	1.60
2	0.89	1.14
3	0.86	1.10
12	0.57	40.52

(a) Which degree would you deploy, and why?

(b) Describe what is happening at degree 12 in terms of the two columns.

(c) Degree 12 has the lowest training error of the four. Explain in three lines why that is not an argument in its favour.

(d) Sketch the shape of both curves against degree and label the region called "overfitting".

U1.18 [⋆⋆⋆] A student runs `SelectKBest(f_classif, k=20)` on all 100 rows of a dataset of 5000 pure-noise features with coin-flip labels, then cross-validates a classifier on the 20 selected columns and reports 0.81.

(a) State the accuracy an honest procedure would report, and why.

(b) Explain the mechanism producing 0.81 in five lines.

(c) State the one-line code change that fixes it.

U1.19 [⋆⋆] Write out, in twelve lines or fewer, a complete scikit-learn workflow that loads a dataset, splits it with stratification and a fixed seed, scales the features inside a pipeline, fits a k-NN classifier and reports test accuracy. Annotate the two lines that prevent leakage.

U1.20 [⋆⋆] Vectorised NumPy was measured at 1.85 ms against 162.9 ms for a Python loop over 10^6 elements. Explain where the 88× comes from, and name one situation where the loop would be the better choice.

§C — Ten-mark questions

U1.21 [⋆⋆] "The goal of machine learning is never to fit the data; it is to generalise." Develop this into a full answer covering: what a training score measures; why a held-out set is needed; what cross-validation adds over a single split; the three-way split and what each part is for; and two distinct mechanisms by which a reported score can be optimistically biased. Use a worked numerical illustration for at least one point.

U1.22 [⋆⋆] You are handed a labelled dataset and asked to "build a model". Set out the complete workflow from receiving the file to reporting a number you would defend, naming the scikit-learn object used at every stage, and stating at each stage what could go wrong and how the pipeline prevents it.

U1.23 [⋆] Compare a rule-based spam filter with a learned one across: how each is built; how each adapts to new spam; how each is debugged; how each is explained to a regulator; and what happens to each when the distribution of email changes. Conclude with the conditions under which you would choose each.

U1.24 [⋆] Explain the estimator/transformer API as a design decision. Why does scikit-learn insist that every object implements the same three or four methods? Discuss what this buys for pipelines, cross-validation and grid search, and give a concrete example of a bug the design makes impossible.

Unit II — Loss functions (L2)

§A — Two-mark questions

U2.1 [⋆⋆⋆] Write the formulae for MSE, MAE and RMSE. State which is most sensitive to outliers and justify it *from the formula.*

U2.2 [⋆⋆⋆] Distinguish a loss, a cost (objective) and a metric. Give one example of a problem where the loss and the metric deliberately differ.

U2.3 [⋆⋆⋆] State the two properties a function must have to serve as a loss.

U2.4 [⋆⋆⋆] Which constant minimises $\sum_i (y_i - c)^2$? Which minimises $\sum_i |y_i - c|$? Which minimises the 0–1 loss?

U2.5 [⋆⋆⋆] Write the Huber loss with parameter δ. State why the second branch is $\delta(|r| - \frac{1}{2}\delta)$ rather than $\delta|r|$.

U2.6 [⋆⋆] State the units of δ in the Huber loss, and one wrong way to choose it.

U2.7 [⋆⋆⋆] Why can accuracy not be used as a training loss? Give two reasons.

U2.8 [⋆⋆⋆] Write the binary cross-entropy formula. State what it becomes for a single example whose predicted probability on the correct class is p.

U2.9 [⋆⋆⋆] Categorical and sparse categorical cross-entropy: state the label format each expects, and whether the two give different numbers.

U2.10 [⋆⋆] Why does the sparse form exist at all? Give the memory argument with a concrete example.

U2.11 [⋆⋆] Why do libraries clip predicted probabilities to $[\varepsilon, 1-\varepsilon]$ before taking logs?

U2.12 [⋆⋆⋆] Define the margin m_i for a binary classifier with $y \in \{-1, +1\}$. Write the hinge loss in terms of it.

U2.13 [⋆⋆⋆] State the one behavioural difference between hinge loss and logistic loss for a point that is already correctly and confidently classified. Say what consequence this has for SVMs.

U2.14 [⋆⋆] Write the triplet loss. Name its three inputs and say what α does.

U2.15 [⋆⋆] Why is random triplet sampling ineffective, and what is the remedy called?

U2.16 [⋆] What is a surrogate loss? Name the two standard surrogates for the 0–1 loss.

U2.17 [⋆⋆] For each, name the loss you would train with: (a) house prices with 2% data-entry errors; (b) 10-class image labels stored as integers; (c) a face-embedding model; (d) a two-class medical screen where you need calibrated probabilities.

§B — Five-mark questions

U2.18 [⋆⋆⋆][**NUM**] A model gives $\hat{y} = (3.5, 5.0, 4.5, 8.0, 5.0)$ against $y = (3, 6, 2, 9, 5)$.

(a) Tabulate the residuals, their squares and their absolute values.

(b) Compute MSE, MAE and RMSE.

(c) Compute the fraction of the total squared error, and of the total absolute error, contributed by the single worst point.

(d) Comment in three lines on what those two fractions tell you.

U2.19 [⋆⋆⋆][**NUM**] Using the residuals of U2.18, compute the Huber loss at $\delta = 1$, $\delta = 1.5$ and $\delta = 3$. For each, state which points took the quadratic branch. Explain in three lines what happens to the Huber loss as δ grows large, and what it converges to.

U2.20 [⋆⋆⋆][**NUM**] Six exam marks are $20, 22, 24, 25, 27, 28$.

(a) Compute the mean and the median.

(b) The last mark is now recorded as 148 by mistake. Recompute both.

(c) State which constant minimises the squared error on the corrupted set, and which minimises the absolute error.

(d) Explain in four lines why “MSE fits the mean, MAE fits the median” is the same statement as (c).

U2.21 [⋆⋆⋆][**NUM**] Four emails have true labels $y = (1, 0, 1, 0)$ and the model predicts $p = (0.8, 0.3, 0.6, 0.1)$ for class 1.

(a) Write down the probability each example assigns to its *correct* class.

(b) Compute $-\log$ of each and the binary cross-entropy.

(c) The third email’s probability now falls to 0.02. Recompute the BCE and state the increase.

(d) Explain in three lines why one confidently wrong prediction can dominate the loss over a large batch.

U2.22 [⋆⋆][**NUM**] A three-class model outputs $P = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.7 & 0.1 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}$ and the true classes are $0, 1, 2$.

(a) Compute the categorical cross-entropy using one-hot labels.

(b) Compute the sparse categorical cross-entropy using integer labels.

(c) State the relationship between your two answers and explain it.

(d) State the memory saved by the sparse form for $K = 10{,}000$ classes and a batch of 256.

U2.23 [⋆⋆⋆][**NUM**] Complete this table and comment on the last column.

p on the correct class	$-\log p$	ratio to the $p = 0.99$ row
0.99		
0.90		
0.50		
0.10		
0.01		

State the exact value of the $p = 0.50$ row and why it is a familiar constant.

U2.24 [⋆⋆⋆][**NUM**] Four points have true labels $y = (+1, -1, +1, -1)$ and scores $f = (1.5, -0.4, 0.2, -0.9)$.

(a) Compute the margin of each point.

(b) Compute the hinge loss of each and the mean.

(c) Compute the squared hinge loss of each and the mean.

(d) Every point is correctly classified. Three of them still cost something. Explain why in four lines, and say what the loss is asking of the model that mere correctness does not.

U2.25 [⋆⋆][**NUM**] Compute the triplet loss for each, taking α as given: (a) $d(a,p) = 0.9$, $d(a,n) = 1.6$, $\alpha = 1.0$; (b) $d(a,p) = 0.4$, $d(a,n) = 2.1$, $\alpha = 1.0$; (c) $d(a,p) = 1.2$, $d(a,n) = 1.4$, $\alpha = 0.5$. For each, state whether the triplet produces a gradient, and explain what case (b) implies about sampling triplets at random.

U2.26 [⋆⋆] A regression model trained with MSE reports RMSE 1.17 and a model trained with MAE reports MAE 1.00 on the same data. A classmate concludes the second model is better. Give a full rebuttal.

U2.27 [⋆⋆] Explain why the MAE is harder to optimise with gradient methods than the MSE. Your answer must mention the gradient of each, the point $r = 0$, and what happens on an even-sized sample.

U2.28 [⋆⋆] A classifier is being trained on 4 classes. Sketch and explain what goes wrong in each case: (a) softmax output with binary cross-entropy; (b) integer labels with categorical cross-entropy; (c) sigmoid output on 4 units with categorical cross-entropy.

U2.29 [⋆] Explain the family portrait of margin losses: the 0–1 loss and three smooth surrogates plotted against the margin. State which are upper bounds on the 0–1 loss, where the hinge loss reaches exactly zero, and how hinge and squared hinge differ for very negative margins.

§C — Ten-mark questions

U2.30 [⋆⋆⋆] "Choosing a loss is not choosing how to score your model; it is choosing what your model will become." Defend this claim in full. Cover: the constant-prediction result for MSE, MAE and 0–1 loss; a worked numerical demonstration on a small contaminated sample; the Huber compromise and the role of δ; and a case where the loss you train with should differ from the metric you report.

U2.31 [⋆⋆⋆] Give a complete account of cross-entropy: the binary form, the categorical form, the sparse categorical form; what each expects of the labels and of the output layer; why the negative log appears; the cost of being confidently wrong with numbers; and why cross-entropy rather than accuracy is used as the training objective.

U2.32 [⋆⋆] Compare MSE, MAE and Huber loss across: definition; sensitivity to outliers; the constant they fit; differentiability; gradient behaviour; hyperparameters; and the situation each is right for. Include one worked numerical comparison on a five-point sample of your own construction.

U2.33 [⋆] Explain margin-based losses in full: the definition of a margin; the hinge loss and its geometry; why it produces support vectors; the squared hinge; the triplet loss and metric learning; and why triplet mining is necessary.

§D — Statements to judge (2 marks each: true or false, *with* a reason)

U2.34 [⋆⋆] "MAE = 1.0 is a better result than MSE = 1.375 on the same predictions."

U2.35 [⋆⋆] "The Huber loss is differentiable everywhere, including at $|r| = \delta$."

U2.36 [⋆⋆] "Categorical and sparse categorical cross-entropy are different loss functions."

U2.37 [⋆⋆] "A point classified correctly always contributes zero hinge loss."

U2.38 [⋆] "RMSE is just MSE in different units, so the two always rank models identically."

Unit III — Optimiser functions (L3, L4)

§A — Two-mark questions

U3.1 [⋆⋆⋆] Write the gradient descent update rule in vector form. Define every symbol.

U3.2 [⋆⋆⋆] Least squares has a closed-form solution. Give two reasons to use gradient descent instead.

U3.3 [⋆⋆⋆] Define *epoch* and *update*. How many updates per epoch does batch gradient descent perform? SGD? Mini-batch with batch size B on n rows?

U3.4 [⋆⋆⋆] State the stability condition on η for a quadratic with largest curvature L. What value of η converges in exactly one step, and what is that method called?

U3.5 [⋆⋆⋆] Write the SGD update. State the property of the per-example gradient that makes SGD work at all.

U3.6 [⋆⋆] State the Robbins–Monro conditions on a step-size schedule and give a schedule satisfying both.

U3.7 [⋆⋆⋆] How does the standard deviation of a mini-batch gradient depend on B? State the practical consequence in one line.

U3.8 [⋆⋆⋆] Write the momentum update rule. State what β controls and what happens as $\beta \to 1$.

U3.9 [⋆⋆⋆] Give the effective step size of momentum in terms of η and β. What does this imply when you raise β from 0.9 to 0.99?

U3.10 [⋆⋆] Define the condition number κ. State the convergence rate of gradient descent and of momentum in terms of it.

U3.11 [⋆⋆] Name a situation in which momentum makes optimisation *slower*, and say why.

U3.12 [⋆⋆⋆] Write the Adagrad update. State the one thing s_t accumulates.

U3.13 [⋆⋆⋆] State Adagrad's flaw on long training runs. Give the rate at which its effective learning rate decays.

U3.14 [⋆⋆⋆] Write the RMSprop update. State the single symbol that differs from Adagrad and what it changes.

U3.15 [⋆⋆] What does RMSprop give up relative to Adagrad?

U3.16 [⋆⋆] State Adadelta's motivating argument in one sentence. What does it remove from the update?

U3.17 [⋆⋆] "ε is Adadelta's hidden learning rate." Explain in three lines.

U3.18 [⋆⋆⋆] Write all four Adam equations.

U3.19 [⋆⋆⋆] Why does Adam need bias correction? Answer in terms of the initialisation of m and v.

U3.20 [⋆⋆⋆] Show that Adam's first corrected step is exactly $\pm\eta$, whatever the size of g_1.

U3.21 [⋆⋆] Adam is not simply "better" than SGD. State what adaptive optimisers actually buy you.

U3.22 [⋆⋆] Name the two distinct reasons a single scalar η can fail. State which one momentum addresses and which it does not.

U3.23 [⋆⋆] Give the four signatures of a loss curve and the action each calls for.

U3.24 [⋆] `SGDRegressor` has no `batch_size` argument. What does that tell you about its implementation?

U3.25 [⋆] Name the two conventions for the momentum update and state the factor by which they differ.

§B — Five-mark questions

U3.26 [⋆⋆⋆][**NUM**] Minimise $f(w) = (w-4)^2$ by gradient descent from $w_0 = 0$ with $\eta = 0.15$.

(a) Write $f'(w)$ and show the update simplifies to $w \leftarrow 0.7w + 1.2$.

(b) Tabulate k, w_k, $f(w_k)$ and $f'(w_k)$ for $k = 0, 1, 2, 3$ and give w_4.

(c) State the ratio by which successive gradients fall, and show it equals $|1 - 2\eta|$.

(d) Give the closed form $w_k = 4 - 4(1-2\eta)^k$ and verify w_4.

U3.27 [⋆⋆⋆][**NUM**] For $f(w) = 3w^2$:

(a) State f'' and the range of η for which gradient descent converges.

(b) Give the η that converges in one step.

(c) For $\eta = 0.4$ and $w_0 = 1$, compute w_1 to w_4 and describe the behaviour.

(d) State what a loss curve would look like in case (c), and the first thing you would do about it.

U3.28 [⋆⋆⋆][**NUM**] Fit $\hat{y} = wx$ by minimising $J(w) = \frac{1}{n}\sum(y_i - wx_i)^2$ on $(1, 3)$, $(2, 5)$, $(4, 9)$, from $w_0 = 0$ with $\eta = 0.05$.

(a) Compute the full-batch gradient at w_0 and the resulting w after one batch update.

(b) Perform one epoch of SGD, visiting the points in the order given, using the per-example gradient $2x_i(wx_i - y_i)$. Report w after each of the three updates.

(c) Compute the exact optimum $w^\star = \sum x_i y_i / \sum x_i^2$.

(d) Compare (a) and (b) against (c) and comment in three lines.

U3.29 [⋆⋆⋆][**NUM**] Repeat U3.26(b) with momentum, $v_t = \beta v_{t-1} + g_t$ and $w_t = w_{t-1} - \eta v_t$, using $\beta = 0.9$, $v_0 = 0$, for three steps. Tabulate g_t, v_t and w_t. Compare w_3 with plain gradient descent's w_3 and with the optimum, and explain what you observe.

U3.30 [⋆⋆][**NUM**] You tuned $\eta = 0.01$ with $\beta = 0.9$ and now wish to use $\beta = 0.99$.

(a) State the effective step size in each case.

(b) Give the rescaling rule and the new η.

(c) Explain in three lines what happens if you raise β without rescaling.

U3.31 [⋆⋆⋆][**NUM**] Perform three steps of **Adagrad** on $f(w) = (w-4)^2$ from $w_0 = 0$ with $\eta = 0.5$, $s_0 = 0$, $\varepsilon = 10^{-8}$. Tabulate t, g_t, s_t, $\sqrt{s_t}$, the step and w_t. Then state, with numbers, the percentage by which the step shrank against the percentage by which the gradient shrank, and explain the difference.

U3.32 [⋆⋆⋆][**NUM**] Repeat U3.31 for **RMSprop** with $\gamma = 0.9$.

(a) Tabulate the same six columns.

(b) Show that the first step equals $\eta/\sqrt{1-\gamma}$ and evaluate it.

(c) At which step does s_t first *fall*, and why can Adagrad's s_t never do this?

U3.33 [⋆⋆⋆][**NUM**] Perform three steps of **Adam** on $f(w) = (w-4)^2$ from $w_0 = 0$ with $\eta = 0.2$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$. Tabulate t, g_t, m_t, v_t, $\hat{m}_t$, $\hat{v}_t$, $\sqrt{\hat{v}_t}$, the step and w_t. Comment on the three step sizes.

U3.34 [⋆⋆⋆][**NUM**] Repeat U3.33 *without* bias correction.

(a) Report the three uncorrected steps.

(b) Compute the ratio $(1-\beta_1)/\sqrt{1-\beta_2}$ and show it matches the first-step discrepancy.

(c) Explain in four lines why the two moments are biased by different factors, and why that mismatch — rather than the shrinkage itself — is the problem.

U3.35 [⋆⋆][**NUM**] Bias correction matters until $1/(1-\beta^t)$ is within 1% of 1.

(a) Derive the condition on t.

(b) Evaluate it for $\beta = 0.9$, 0.99 and 0.999.

(c) Comment on what your answer for $\beta_2 = 0.999$ implies for a training run of 2000 steps.

U3.36 [⋆⋆][**NUM**] Adadelta's first step is $\Delta w_1 = -\frac{\sqrt{d_0 + \varepsilon}}{\sqrt{s_1 + \varepsilon}} g_1$ with $d_0 = 0$ and $s_1 = (1 - \rho)g_1^2$. Take $g_1 = -8$, $\rho = 0.9$. Compute Δw_1 for $\varepsilon = 10^{-6}$, 10^{-8} and 10^{-12}, and state the relationship you observe. Explain what this means for the claim that Adadelta has no learning rate.

U3.37 [⋆⋆][**NUM**] Complete the table and answer the question below it.

B	relative gradient noise $1/\sqrt{B}$	work per update
1		
4		
16		
64		
256		

Doubling B costs twice the work. By what factor does it reduce the noise? State the practical conclusion this leads to about the choice of B.

U3.38 [⋆⋆] Complete the optimiser summary table and add one line of comment under each row.

Method	B	**Updates/epoch**	**Noise**	**Cost/update**
Batch				
Mini-batch				
Stochastic				

U3.39 [⋆ ⋆ ⋆] Write from memory the update rules for SGD, momentum, Adagrad, RMSprop and Adam, in that order. For each, state in one line what it added over its predecessor.

U3.40 [⋆⋆] "Every optimiser in this lecture is the line $w_i \leftarrow w_i - \frac{\eta}{\sqrt{s_i}} g_i$; they differ only in what s_i accumulates." Justify this for Adagrad, RMSprop and Adam, stating s_i in each case.

U3.41 [⋆⋆] Give the dimensional-analysis argument for Adadelta. What are the units of a plain SGD step, of an Adagrad step, and of a Newton step? Which is dimensionally correct, and why is it not used?

U3.42 [⋆⋆] A model has 20 dense features and 20 very rare ones. Explain why a single learning rate cannot serve both, name the optimiser you would reach for, and justify it in terms of what that optimiser accumulates.

U3.43 [⋆] Explain why standardising features is an *optimisation* decision, not merely a preprocessing one. Refer to the condition number and to the stability limit $2/L$.

U3.44 [⋆⋆] Give the six-step recipe for training a model with gradient methods, from raw features to a final learning-rate decay.

§C — Ten-mark questions

U3.45 [⋆ ⋆ ⋆] Give a complete account of the learning rate: the update rule; the four regimes of η with the multiplier $|1 - \eta f''|$ in each; the stability threshold $2/L$ and its derivation for a quadratic; the four loss curve signatures; and how you would find a good η in practice. Include a worked numerical example on a one-dimensional quadratic.

U3.46 [$\star\star\star$] Trace the development from batch gradient descent to Adam as a chain of fixes, at each step naming the problem, the fix, the update rule and what the fix cost. Cover batch $\to$ SGD $\to$ mini-batch $\to$ momentum $\to$ Adagrad $\to$ RMSprop $\to$ Adam.

U3.47 [$\star\star\star$] Explain momentum completely: the ravine problem; the heavy-ball rule; the exponentially weighted average interpretation and the unrolled form; the effective step $\eta/(1-\beta)$; the modified stability ceiling; overshoot; the improvement in convergence rate from κ to $\sqrt{\kappa}$; and when momentum does not help.

U3.48 [$\star\star\star$] Explain Adam in full: the two moments and what each is for; the bias-correction derivation for a constant gradient; the first-step result; the size of the error if correction is omitted; the defaults and what each controls; and the most common misconception about Adam.

U3.49 [$\star\star$] Compare Adagrad, RMSprop, Adadelta and Adam across: what is accumulated; whether it decays; the size of the first step; the hyperparameters; the problem each was invented to fix; and the situation each is right for. Support with at least two hand-computed steps of any two of them.

U3.50 [$\star\star$] "The real benefit of an adaptive optimiser is not a better final answer but a wider window of learning rates that work." Explain and defend this, and describe the experiment you would run to demonstrate it.

§D — Statements to judge (2 marks each)

U3.51 [$\star\star$] "Adam is always a better choice than SGD with momentum."

U3.52 [$\star\star$] "Because Adam is adaptive, it does not need a learning-rate schedule."

Unit IV — Data preprocessing (L5, L6, L7)

§A — Two-mark questions

U4.1 [$\star\star\star$] Name Han, Kamber and Pei's four preprocessing tasks.

U4.2 [$\star\star\star$] Give the probability that a row survives listwise deletion when each of d columns is independently missing with probability p. Evaluate it for $p = 0.05$, $d = 30$.

U4.3 [$\star\star\star$] Name three imputation strategies for a numeric column and one for a categorical column, with the situation each suits.

U4.4 [$\star\star\star$] State the factor by which mean imputation shrinks the standard deviation of a column, in terms of the fraction observed.

U4.5 [$\star\star\star$] Define MCAR, MAR and MNAR. Give one example of each.

U4.6 [$\star\star$] For which of the three mechanisms can a multivariate imputer recover the bias, and for which can nothing in the data help?

U4.7 [$\star\star$] What is a missingness indicator, and when is adding one leakage?

U4.8 [$\star\star\star$] Write the z-score rule, the Tukey fence rule and the modified z-score rule for outlier detection.

U4.9 [$\star\star\star$] Give the breakdown point of each of the three outlier rules.

U4.10 [$\star\star\star$] Define *masking*. Explain in two lines why it defeats the z-score rule.

U4.11 [$\star\star$] State the largest $|z|$ attainable in a sample of size n. For what n does the $|z| > 3$ rule become capable of firing at all?

U4.12 [$\star\star$] Why is 0.6745 the constant in the modified z-score? Why is the fence multiplier 1.5?

U4.13 [$\star\star$] Distinguish skew from contamination. State the order in which you should transform and detect outliers, with a reason.

U4.14 [$\star\star$] Name four ways to dispose of an identified outlier, and state when each is appropriate.

U4.15 [$\star$] Distinguish equal-width from equal-depth binning. Name the three smoothing methods.

U4.16 [$\star\star\star$] Write the Box–Cox transformation. State its one restriction on the data and name the transformation that removes it.

U4.17 [$\star\star$] Why does the Box–Cox formula subtract 1 and divide by λ?

U4.18 [$\star\star$] How is λ chosen in practice? State what it is *not* chosen to do.

U4.19 [$\star\star\star$] Name the four scalers, write each formula and give each output range.

U4.20 [$\star\star\star$] Which of the four scalers is completely destroyed by a single extreme value, and why? Which is the safest?

U4.21 [$\star\star\star$] Why does scaling not change a decision tree's predictions? Give the argument from the form of a split.

U4.22 [$\star\star$] Distinguish scaling from transformation. Which one changes skewness?

U4.23 [$\star\star\star$] Distinguish nominal from ordinal variables. State the encoding each requires.

U4.24 [$\star\star\star$] What is the dummy variable trap? State what it breaks and what it does *not* break.

U4.25 [$\star\star$] State the default ordering `OrdinalEncoder` applies, and why that is a dangerous default.

U4.26 [$\star\star$] What does `handle_unknown='ignore'` do, and what happens without it?

U4.27 [$\star\star$] Name Han, Kamber and Pei's three kinds of data reduction.

U4.28 [⋆⋆⋆] State the curse of dimensionality in terms of nearest and farthest distances. Name the class of method it damages.

U4.29 [⋆⋆⋆] Give the four steps of PCA.

U4.30 [⋆⋆⋆] What does the jth eigenvalue of the covariance matrix equal? What does the trace equal?

U4.31 [⋆⋆⋆] Why must you standardise before PCA? Give the failure mode concretely.

U4.32 [⋆⋆⋆] "PCA reduces dimensionality by keeping the most important features." State precisely what is wrong with this sentence.

U4.33 [⋆⋆] PCA has never seen the labels. Give a concrete case where this makes it the wrong tool, and name a supervised alternative.

U4.34 [⋆⋆⋆] Name the three families of feature selection with one method from each, and state the cost and risk of each family.

U4.35 [⋆⋆] The ANOVA F test and mutual information rank features differently. State what each measures and construct a feature that one finds and the other misses.

U4.36 [⋆⋆] "L1 selects, L2 shrinks." Explain.

U4.37 [⋆⋆⋆] State the single rule that determines what may be fitted on the full dataset and what may not.

U4.38 [⋆⋆⋆] Which leaks more: fitting a scaler on all the data, or performing feature selection on all the data? Give the reason.

U4.39 [⋆⋆⋆] Why must a test set be stratified for classification? Name the scikit-learn argument.

U4.40 [⋆⋆⋆] State the default value of `shuffle` in `KFold` and the failure it causes on a sorted file.

U4.41 [⋆⋆⋆] What is the "unit of independence"? Give three examples where it is not the row.

U4.42 [⋆⋆⋆] Name the splitter for each: repeated measurements per patient; time-ordered rows; imbalanced classes; imbalanced classes *and* repeated measurements.

U4.43 [⋆⋆] Give the k-fold cost model: number of fits, rows per fit, and total row-fits.

U4.44 [⋆⋆] Why is LOOCV rarely worth it? Give both the cost and the variance argument.

U4.45 [⋆⋆] Why is `cross_val_score(...).std()` not the uncertainty of the mean?

U4.46 [⋆⋆⋆] Distinguish the roles of a validation set and a test set. How many times may you look at each?

U4.47 [⋆⋆] Why is `GridSearchCV.best_score_` not a performance estimate? What is?

U4.48 [$\star\star\star$] Write the formulae for accuracy, precision, recall, F_1 and balanced accuracy from a confusion matrix.

U4.49 [$\star\star\star$] What is the chance baseline of ROC–AUC? Of PR–AUC?

U4.50 [$\star\star\star$] Write the SMOTE interpolation formula and define every symbol.

U4.51 [$\star\star$] Give two situations in which SMOTE is the wrong tool.

U4.52 [$\star\star\star$] Where in a cross-validation loop must resampling happen? State what goes wrong if it happens earlier.

U4.53 [$\star\star$] Name a cheaper alternative to resampling that achieves the same effect on recall, and state two further advantages it has.

§B — Five-mark questions

U4.54 [$\star\star\star$][**NUM**] Eight students sat a test; two scripts were lost. The six recorded marks are $48, 55, 60, 67, 72, 78$.

(a) Compute the mean, median and standard deviation of the observed marks (use ddof $= 1$ for the sd, and state the population sd too).

(b) Impute the two missing marks with the mean. Recompute the mean and the population standard deviation of all eight.

(c) Repeat with median imputation.

(d) Compute $\sqrt{m/n}$ and compare it with the ratio of the imputed sd to the observed sd. Explain the agreement.

U4.55 [$\star\star\star$][**NUM**] Complete the survival table for listwise deletion.

p	$d = 5$	$d = 20$	$d = 30$	$d = 50$
0.01				
0.05				
0.10				

(a) Fill in all twelve cells.

(b) At $p = 0.05$, find the number of columns at which half the rows are lost.

(c) At $p = 0.10$ and $d = 50$, how many raw rows must you collect to retain 1000 complete ones?

(d) State in three lines when listwise deletion is nevertheless the right choice.

U4.56 [$\star\star\star$][**NUM**] A sensor column reads 20, 21, 19, 22, 20, 21, 23, 19, 20, 22, 120, 125.

(a) Compute the mean, the sample standard deviation, the median and the MAD.

(b) Compute Q_1, Q_3, the IQR and the 1.5×IQR fences.

(c) Score the value 120 under all three rules: $|z| > 3$, the Tukey fence, and the modified z at threshold 3.5. State which rules fire.

(d) Compute the standard deviation of the ten genuine readings alone, and the z-score the value 120 would then have. Explain the discrepancy with (c) and name the effect.

U4.57 [$\star\star\star$][**NUM**] Complete the table and answer the question.

n	largest attainable $\lvert z\rvert$	can $\lvert z\rvert > 3$ ever fire?
5		
8		
10		
11		
20		

Derive the bound $|z_1| \leq (n-1)/\sqrt{n}$. State what it means for a practitioner who writes `if abs(z) > 3` in a script that processes batches of eight readings.

U4.58 [$\star\star$][**NUM**] Nine prices are $5, 8, 14, 17, 20, 26, 27, 30, 36$.

(a) Partition them into three equal-depth bins and give each bin's mean, median and boundaries.

(b) Write the column smoothed by bin means, by bin medians and by bin boundaries.

(c) State the bin widths and bin sizes that equal-width binning would have produced instead.

(d) State one thing binning gains and one thing it costs.

U4.59 [$\star\star$][**NUM**] Using the sensor column of U4.56, apply each treatment and report the resulting mean and standard deviation: (a) leave it alone; (b) winsorise at the 10th and 90th percentiles; (c) cap at the IQR fences; (d) drop the points flagged by the modified z. Rank the four by how close they come to the truth (mean 20.7, sd 1.34) and justify your ranking.

U4.60 [$\star\star$][**NUM**] The values $2, 6, 18, 54$ are in geometric progression.

(a) Compute the gaps between consecutive values.

(b) Compute the natural logarithm of each and the gaps between those.

(c) State the exact value the log gaps take and why.

(d) Explain in three lines why this makes the log the useful "middle rung" of the ladder of powers.

U4.61 [$\star\star$][**NUM**] Apply the Box–Cox formula $(x^\lambda - 1)/\lambda$ to $x = 1, 4, 16, 64$ for $\lambda = 0.5$, -0.5 and 0.001. Compare the $\lambda = 0.001$ column with $\log x$ and explain what you observe.

U4.62 [$\star\star\star$][**NUM**] The values $3, 5, 5, 5, 6, 6, 8, 10$.

(a) Compute the mean, population sd, min, max, median, Q_1, Q_3 and IQR.

(b) Transform the column with `StandardScaler`, `MinMaxScaler`, `RobustScaler` and `MaxAbsScaler`, giving all eight values in each case.

(c) State the output range each scaler guarantees.

U4.63 [$\star\star\star$][**NUM**] A typing error appends the value 400 to the column of U4.62.

(a) Recompute the mean, sd, median and IQR.

(b) For `StandardScaler`, `MinMaxScaler` and `RobustScaler`, compute the span the eight *good* values now occupy, and the factor by which each has been compressed.

(c) Name the scaler that survives, and explain why in terms of the statistics each one uses.

(d) Explain in three lines why `MinMaxScaler` is described as "the dangerous one" despite producing output that looks correct.

U4.64 [$\star\star\star$][**NUM**] Three customers are recorded as (age, annual income in rupees): $A = (30,\ 500\,000)$, $B = (55,\ 505\,000)$, $C = (31,\ 900\,000)$.

(a) Compute $d(A, B)^2$ and $d(A, C)^2$ on the raw values, showing each coordinate's contribution and the percentage income supplies.

(b) Standardise all three columns (population sd) and recompute both squared distances.

(c) State A's nearest neighbour before and after.

(d) Explain in four lines what "choosing not to scale" actually chooses.

U4.65 [$\star\star\star$][**NUM**] A nominal colour variable is ordinal-encoded alphabetically: blue $= 0$, green $= 1$, red $= 2$. The true group means of the target are blue 30, green 70, red 15.

(a) Compute $\bar{x}$, $\bar{y}$, S_{xy} and S_{xx}.

(b) Fit the least-squares line, giving slope and intercept.

(c) Give the prediction for each colour, the SSE, the SST and R^2.

(d) State the R^2 a one-hot encoding would achieve and explain the difference in four lines.

U4.66 [$\star\star$][**NUM**] A categorical column has 5 levels.

(a) How many columns does one-hot encoding produce, with and without `drop='first'`?

(b) A design matrix has an intercept column and all 5 dummies. State its rank and why.

(c) State what rank deficiency breaks, and what it does *not* break.

(d) State when you should use `drop='first'` and when you should not.

U4.67 [⋆⋆][**NUM**] `PolynomialFeatures` generates $\binom{d+k}{k} - 1$ features. Evaluate for $(d, k) = (2, 2), (10, 2), (30, 2), (10, 3), (100, 2)$. Comment on what this implies for "just generate all the interactions".

U4.68 [⋆⋆⋆][**NUM**] Perform PCA by hand on the five points $(1, 1), (2, 3), (3, 2), (4, 5), (5, 4)$.

(a) Compute the mean and the centred matrix.

(b) Compute the 2×2 covariance matrix with divisor $n - 1$.

(c) Find both eigenvalues and the first eigenvector. Verify the trace.

(d) Give the proportion of variance explained by each component.

(e) Project onto PC1, give the five scores, and verify their variance equals λ_1.

(f) Reconstruct from one component and compute the total squared reconstruction error. Verify it equals $(n - 1)\lambda_2$.

U4.69 [⋆⋆][**NUM**] A PCA on six standardised features gives eigenvalues $4.8,\ 2.4,\ 1.3,\ 0.8,\ 0.4,\ 0.3$.

(a) Give the proportion of variance explained by each component.

(b) Give the cumulative proportions.

(c) State the smallest k reaching 80%, 90% and 95%.

(d) State the relative reconstruction error at each of those three k, without further computation, and justify why no computation is needed.

U4.70 [⋆ ⋆ ⋆][**NUM**] Twelve rows contain three positives, all at the end of the file. Four rows are held out as a test set.

(a) If the last four rows are taken, how many positives are in the test set and how many in training? State the consequence.

(b) If the rows are shuffled first, compute the probability that the test set contains *no* positive, as a ratio of binomial coefficients.

(c) State exactly what `stratify=y` produces.

(d) State what `recall` evaluates to on a fold containing no positives, and what scikit-learn does about it.

U4.71 [⋆ ⋆ ⋆][**NUM**] A screening model on 10,000 patients gives TN = 9900, FP = 60, FN = 30, TP = 10.

(a) Compute accuracy, precision, recall, F_1 and balanced accuracy.

(b) Compute the accuracy of a model that predicts "negative" always.

(c) State which of the two models scores higher on accuracy, and what that tells you about the metric.

(d) Name the metric you would report to a clinical committee and justify it in three lines.

U4.72 [$\star\star\star$][**NUM**] Complete the cross-validation cost table for $n = 5000$.

Scheme	Fits	Rows per fit	Row-fits
2-fold			
5-fold			
10-fold			
LOOCV			

If one fit on 4000 rows takes 0.4 seconds and cost is linear in rows, give the wall-clock time of each scheme. State which you would use and why.

U4.73 [$\star\star$][**NUM**] A dataset has 5000 rows and an 8% positive rate, and you run 5-fold stratified cross-validation.

(a) Give the number of rows and positives in each validation fold and in each training portion.

(b) The training portion is randomly oversampled to balance. Give the resulting row count and class counts, and the percentage of minority rows that are duplicates.

(c) Repeat for random undersampling, and state the percentage of majority rows discarded.

(d) State what happens to the validation fold in each case, and why.

U4.74 [$\star\star$][**NUM**] Two minority patients are at $A = (2, 3)$ and $B = (6, 11)$.

(a) Compute the SMOTE synthetic point for $u = 0,\ 0.25,\ 0.5,\ 0.75,\ 1$.

(b) State the geometric object all synthetic points lie on.

(c) State the assumption SMOTE is making, in one sentence.

(d) Give one dataset for which that assumption is clearly false.

U4.75 [$\star\star$][**NUM**] A model reports accuracy 0.91 on a test set of 44 rows.

(a) Compute the standard error of that accuracy.

(b) State how much one row is worth in accuracy points.

(c) Compute the test-set size needed to halve the standard error.

(d) Explain in three lines why comparing two models on one 44-row test set is unsound.

U4.76 [$\star\star\star$] A student oversamples the minority class, then splits 80/20, and reports $F_1 = 0.89$. Doing it the other way round gives $F_1 = 0.11$.

(a) Name the error.

(b) Explain the mechanism in five lines, referring to what appears in both the training and test sets.

(c) State what fraction of the minority test rows are likely to be exact copies of training rows, and why.

(d) State the one-line fix.

§C — Ten-mark questions

U4.77 [$\star\star\star$] You receive 5000 patient records: 12 numeric features (two with about 4% missing values and visible outliers), 3 categorical features, and a binary target that is 8% positive. Design and defend a complete pipeline. Cover: the order of every preprocessing step and why it sits where it does; exactly which objects are fitted on training data only; what happens inside one fold of 5-fold stratified cross-validation, with fold sizes and positive counts; where resampling goes and what breaks if it goes earlier; and which single metric you would report, with a justification in terms of the clinical cost of a false negative.

U4.78 [$\star\star\star$] Give a complete account of missing data: listwise deletion and its exponential cost; the three simple imputers; the variance-shrinkage result with its derivation; MCAR, MAR and MNAR with an example of each; which mechanisms a multivariate imputer can repair; the missingness indicator and when it is itself leakage; and a recommendation with the reason it is not the choice of imputer that matters most.

U4.79 [$\star\star\star$] Give a complete account of outlier detection and treatment: the three rules with their formulae; their breakdown points; masking and swamping, with a worked numerical demonstration; the z-score ceiling and its proof; why skew is not contamination and what to do about it; and the four disposal options with the circumstances for each. Conclude with the distinction between detection and disposal.

U4.80 [$\star\star\star$] Explain feature scaling completely: the three mechanisms by which units matter; the four scalers with formulae and output ranges; a worked numerical demonstration that a nearest neighbour can change; which model families are affected and which are provably not, with the argument; what a single typo does to each scaler; and why scaling is fitted on the training split.

U4.81 [$\star\star\star$] Explain PCA completely: the question it asks; the four-step recipe; the meaning of the eigenvalues and of the trace; a worked by-hand example on five points including the reconstruction error identity; how k is chosen and why there is no universal rule; why standardising comes first; why PCA is not feature selection; and why being unsupervised can make it the wrong tool.

U4.82 [⋆⋆⋆] Explain cross-validation completely: what a single held-out score estimates and how noisy it is; the k-fold recipe and its cost model; how k trades bias against variance against time; why LOOCV's across-fold standard deviation is not what it appears to be; the three failure modes requiring stratified, grouped and time-ordered splitters; and how tuning destroys the estimate unless nested.

U4.83 [⋆⋆⋆] Explain imbalanced data completely: why accuracy fails, with a worked confusion matrix; the metrics that do not fail and their chance baselines; the four available remedies; SMOTE's arithmetic and its assumption; the empirical result that resampling improves recall while costing precision; the threshold-moving alternative; and the rule about where resampling belongs, with the size of the error when it is broken.

U4.84 [⋆⋆⋆] "Anything you learn from the data — a mean, a category list, a feature subset, a hyperparameter, a resampled row — is part of the model, and belongs inside the loop." Take this as a thesis and defend it. Give at least four distinct leaks, ranked by the size of the error each produces, and say for each what the correct construction is.

U4.85 [⋆⋆] Explain feature selection: the three families with two methods each; what a filter measures and why choosing one is choosing an assumption; the disagreement between the ANOVA F test and mutual information; recursive feature elimination and its cost; how L1 differs from L2; why four selectors can disagree almost completely; and why "select features because it is good practice" is bad advice.

U4.86 [⋆⋆] Explain feature engineering: what it is; the four moves; a worked example where one constructed column changes R^2 from 0.66 to 1.00; why a representation is good or bad only relative to a model, with the binning result; and why automatic feature generation is not a substitute for a reason.

§D — Statements to judge (2 marks each)

U4.87 [⋆⋆⋆] "The z-score rule detects outliers more than three standard deviations from the mean."

U4.88 [⋆⋆⋆] "PCA reduces dimensionality by keeping the most important features."

U4.89 [⋆⋆] "Scaling the inputs to a decision tree improves its accuracy."

U4.90 [⋆⋆] "Mean imputation is safe because it does not change the mean."

U4.91 [⋆⋆] "One-hot encoding a variable with 257 levels is fine; the model will ignore the useless columns."

U4.92 [⋆⋆⋆] "Fitting the scaler on the whole dataset is a serious leak that will badly inflate your score."

U4.93 [⋆⋆] "LOOCV is the most thorough form of cross-validation and should be used whenever it is affordable."

U4.94 [⋆⋆⋆] "SMOTE generates new data."

U4.95 [⋆⋆] "A model with 99% accuracy on a 1%-positive problem is performing well."

U4.96 [⋆] "`random_state=42` is a formality."

Unit V (part) — Regression (L8, L9, L10)

This unit is covered only as far as L10

Unit V runs from L8 to L12. The mid-semester paper reaches only **L8, L9 and L10** — regression foundations and least squares, maximum likelihood and model adequacy, and logistic regression. **Multiple linear regression (L11) and regularization (L12) are not in this bank** and are not expected on the paper.
One consequence worth knowing: L10's separation problem is *fixed* by regularization, which is L12. So a question here may ask you to *diagnose* perfect separation and *name* the remedy, but not to work with ridge or lasso.

§A — Two-mark questions

U5.1 [⋆⋆⋆] Write the simple linear regression model, naming every symbol. State which symbols are Greek and which are Roman, and why the distinction matters.

U5.2 [⋆⋆⋆] "Linear" in linear regression means linear in *what*? Classify each as linear or not: (a) $y = \beta_0 + \beta_1 x + \beta_2 x^2$; (b) $y = \beta_0 + \beta_0\beta_1 x$; (c) $y = \beta_0 + \beta_1 \log x$; (d) $y = \beta_0 e^{\beta_1 x}$.

U5.3 [⋆⋆⋆] State the criterion least squares minimises. Write it out in full, and say precisely which deviations are being squared.

U5.4 [⋆⋆⋆] Write the two normal equations for simple linear regression.

U5.5 [⋆⋆⋆] Give the least-squares estimators b_1 and b_0 in terms of S_{xx} and S_{xy}, and define both.

U5.6 [⋆⋆] Give three reasons for choosing squared deviations over absolute deviations, and one thing you give up.

U5.7 [⋆⋆] Why vertical deviations rather than perpendicular ones? Name what you get if you minimise perpendicular distance instead.

U5.8 [⋆⋆⋆] State the second-order condition for the least-squares solution. Give the Hessian and the exact condition under which the solution exists and is unique.

U5.9 [⋆⋆] Give a dataset on which the least-squares line does not exist, and say which quantity is zero.

U5.10 [⋆⋆⋆] Write the matrix form of the normal equations and its solution. State the condition on X.

U5.11 [⋆⋆⋆] State the four properties of the least-squares line. Which one requires an intercept?

U5.12 [⋆⋆] Prove that the residuals sum to zero, from the normal equations.

U5.13 [⋆⋆] Prove that the line passes through $(\bar{x}, \bar{y})$.

U5.14 [⋆⋆⋆] Write the ANOVA decomposition of the total sum of squares and name each term.

U5.15 [⋆⋆⋆] Give $\text{Var}(b_1)$. State what a designer of an experiment should therefore do when choosing the x values.

U5.16 [⋆⋆] Define the leverage h_i. Give the rule for flagging a high-leverage point.

U5.17 [⋆⋆] Compare the closed form with gradient descent for fitting a linear model, across cost, memory, tuning and when each fails.

U5.18 [⋆⋆⋆] State the probability model behind maximum likelihood estimation for regression. What exactly is assumed about ε_i?

U5.19 [⋆⋆⋆] Write the likelihood and the log-likelihood for the Gaussian linear model.

U5.20 [⋆⋆⋆] State in one sentence the central result linking maximum likelihood to least squares.

U5.21 [⋆⋆⋆] Why does the value of σ^2 not affect which β_0, β_1 maximise the likelihood?

U5.22 [⋆⋆] Give the maximum likelihood estimate of σ^2, and the unbiased alternative. State the reason for the difference — *not* by restating the formula.

U5.23 [⋆⋆] Change the noise model from Gaussian to Laplace. What criterion does maximum likelihood then minimise?

U5.24 [⋆⋆⋆] Define R^2 two ways. State its range on training data, and why that range does not apply to held-out data.

U5.25 [⋆⋆⋆] Why can R^2 never decrease when a predictor is added? Answer in one sentence.

U5.26 [⋆⋆] Write adjusted R^2. State what it is and what it is not.

U5.27 [⋆⋆⋆] "$R^2 = 0.85$, so the model is good." State two distinct things wrong with that inference.

U5.28 [⋆⋆] Name the four assumptions of the linear model. For each, state what its violation invalidates.

U5.29 [⋆⋆] Define over-fitting. State why it is a property of the model *and* the data, not the model alone.

U5.30 [⋆⋆⋆] Why does linear regression fail on a 0/1 target? Give two distinct failures.

U5.31 [⋆⋆⋆] Define odds and log-odds. Give the range of each, and say which one $w^\top x + b$ can safely produce.

U5.32 [⋆⋆⋆] Write the logistic regression model in log-odds form, then invert it to get $p(x)$.

U5.33 [⋆⋆⋆] State three properties of the sigmoid, including its derivative.

U5.34 [⋆⋆⋆] Write the log-likelihood for logistic regression. State its relationship to binary cross-entropy.

U5.35 [⋆⋆⋆] Write the gradient of the logistic log-likelihood in matrix form. State in words what it is measuring.

U5.36 [⋆⋆⋆] Why is there no closed form for logistic regression? Contrast the score equations with the normal equations.

U5.37 [⋆⋆] Write the Hessian of the logistic log-likelihood and state what its definiteness buys you — and what it does not.

U5.38 [⋆⋆⋆] Interpret a logistic coefficient β_j. State exactly what e^{β_j} is a change in.

U5.39 [⋆⋆⋆] State the divide-by-four rule and what it bounds.

U5.40 [⋆⋆] Why is the decision threshold not part of the model? Who should choose it?

U5.41 [⋆⋆] Write the softmax. Show that it reduces to the sigmoid when $K = 2$.

U5.42 [⋆⋆⋆] What is perfect separation? State what happens to the maximum likelihood estimate, and name the fix.

U5.43 [⋆] Why is squared error the wrong loss for logistic regression? Give two reasons.

§B — Five-mark questions

U5.44 [⋆⋆⋆][**NUM**] Fit a least-squares line to $x = (0, 1, 2, 3, 4)$, $y = (1, 2, 4, 5, 8)$.

(a) Tabulate $x_i - \bar{x}$, $y_i - \bar{y}$, $(x_i - \bar{x})^2$ and $(x_i - \bar{x})(y_i - \bar{y})$, and give S_{xx}, S_{xy}, S_{yy}.

(b) Compute b_1 and b_0 and write the fitted line.

(c) Give the fitted values and residuals. Verify $\sum e_i = 0$ and $\sum x_i e_i = 0$.

(d) Compute SSE, SSR and SST, and verify the decomposition. Check SSR two independent ways.

(e) Predict y at $x = 6$.

U5.45 [⋆⋆⋆][**NUM**] Repeat the fit of U5.44 by the matrix route.

(a) Write the design matrix X and compute $X^\top X$ and $X^\top y$ from the raw sums.

(b) Compute $\det(X^\top X)$ and verify it equals nS_{xx}.

(c) Invert the 2×2 matrix and solve for b.

(d) State the exact condition under which this route fails, and give a five-row dataset on which it does.

U5.46 [⋆⋆⋆] Derive the least-squares estimators from a blank page. Your answer must contain: the object being minimised; both partial derivatives; both normal equations; the substitution that isolates b_1; the identification of S_{xx} and S_{xy}; and the second-order condition. *Stopping at the first-order conditions is not a complete answer.*

U5.47 [$\star\star\star$][**NUM**] Four candidate lines are proposed for the data of U5.44.

A: $\hat{y} = 0.5 + 1.7x$	B: $\hat{y} = 0.6 + 1.7x$	C: $\hat{y} = 0.6 + 1.8x$	D: $\hat{y} = 1.0 + 1.5x$

For each, compute SSE, $\sum e_i$ and $\sum x_i e_i$. Identify the least-squares line **using both normal equations**, and explain why one of them alone is not enough — naming the candidate that proves it.

U5.48 [$\star\star$][**NUM**] Refit the data of U5.44 *without* an intercept.

(a) Derive the estimator $b_1 = \sum x_i y_i / \sum x_i^2$ from scratch.

(b) Evaluate it.

(c) Give the residuals and their sum.

(d) Property 1 has died. Explain exactly which normal equation is missing and why.

(e) Compare the two SSEs and say whether the comparison is fair.

U5.49 [$\star\star\star$][**NUM**] For the data of U5.44:

(a) Compute the leverage $h_i = 1/n + (x_i - \bar{x})^2/S_{xx}$ of every point, and verify the leverages sum to 2.

(b) Adding δ to y_i changes b_1 by $\delta(x_i - \bar{x})/S_{xx}$. Derive this.

(c) Adding 3 to one y value: give the new b_1 for each of the five choices.

(d) One point cannot move the slope at all. Say which and why.

U5.50 [$\star\star$][**NUM**] $\mathrm{Var}(b_1) = \sigma^2/S_{xx}$ with $\sigma^2 = 4$. Three designs, all with the same budget: all with $n = 5$: equally spaced x over $[0, 8]$; five points clustered on $[3, 5]$; and three points at $x = 0$ with two at $x = 8$. Compute S_{xx}, $\mathrm{Var}(b_1)$ and $\mathrm{sd}(b_1)$ for each. Rank them, then state the one thing the best design cannot do.

U5.51 [$\star\star\star$] Derive the maximum likelihood estimators for the Gaussian linear model from a blank page: the probability model, the one-row density, the likelihood, the log-likelihood, and the argument that maximising it is identical to minimising SSE for *every* $\sigma^2 > 0$.

U5.52 [$\star\star\star$][**NUM**] Using the fit of U5.44 (SSE $= 1.10$, $n = 5$):

(a) Compute $\hat{\sigma}^2_{\mathrm{ML}} = \mathrm{SSE}/n$.

(b) Compute the unbiased estimate $s^2 = \mathrm{SSE}/(n-2)$ and the standard error of the estimate s.

(c) Give the bias factor and the percentage by which the ML estimate understates σ^2 in expectation.

(d) Explain *why* the divisor is $n-2$ — an answer that restates the formula earns nothing.

U5.53 [⋆⋆][**NUM**] For each of $(n, \mathrm{SSE}) = (8, 20)$, $(6, 12)$, $(12, 45)$, compute $\hat{\sigma}^2_{\mathrm{ML}}$, s^2, the bias factor and the percentage understatement. Then find the smallest n for which the understatement falls below 1%.

U5.54 [⋆⋆⋆][**NUM**] For the fit of U5.44:

(a) Compute $\mathrm{SS}_{\mathrm{tot}}$, $\mathrm{SS}_{\mathrm{res}}$ and $\mathrm{SS}_{\mathrm{reg}}$.

(b) Compute R^2 from the definition.

(c) Verify R^2 two further independent ways.

(d) Compute r and confirm $r^2 = R^2$.

(e) State one thing this R^2 does *not* tell you, and what you would look at instead.

U5.55 [⋆⋆][**NUM**] With $n = 25$ and $R^2 = 0.55$ held fixed, compute adjusted R^2 for $p = 1, 5, 10, 15$. Find the smallest p at which adjusted R^2 turns negative, showing the inequality. Explain what a negative adjusted R^2 is telling you.

U5.56 [⋆⋆⋆][**NUM**] On a held-out set, $y = (2, 4, 6)$ and the model predicts $(7, 4, 1)$.

(a) Compute $\mathrm{SS}_{\mathrm{tot}}$, $\mathrm{SS}_{\mathrm{res}}$ and R^2.

(b) Compute R^2 for a model that predicts the constant 4.

(c) State the lower bound on held-out R^2.

(d) Explain in four lines why $R^2 \in [0, 1]$ is a training-set statement only.

U5.57 [⋆⋆][**NUM**] For $x = (1, 2, 3, 4, 5)$, $y = (2, 3, 6, 5, 9)$, compute S_{xx}, S_{xy}, S_{yy}, the direct slope S_{xy}/S_{xx} and the inverse slope S_{yy}/S_{xy}. Show that their ratio equals r^2 and evaluate it. State which line you would report and why, and what goes wrong if you invert the direct fit to predict x from y.

U5.58 [⋆⋆⋆] A student reports a degree-3 polynomial on 331 rows with training $R^2 = 0.918$, and concludes the model is excellent. The cross-validated R^2 is -1212.18 and the held-out R^2 is -22.04; predicting the training mean scores -0.0001.

(a) Name every fault in the reasoning.

(b) State how many parameters a degree-3 expansion of 10 features has, and compare it with the row count.

(c) Explain what a held-out R^2 of -22 means in plain words.

(d) State what should have been reported instead.

U5.59 [⋆⋆⋆][**NUM**] Six tumour sizes $x = (1, 2, 3, 4, 5, 6)$ mm carry labels $y = (0, 0, 0, 1, 1, 1)$.

(a) Fit a least-squares line. Give S_{xx}, S_{xy}, the slope and the intercept.

(b) Give the six predictions. How many lie outside $[0, 1]$?

(c) Give the x at which the prediction crosses 0.5.

(d) A malignant tumour at $x = 20$ mm is added. Refit and give the new boundary. State how far it moved and which original point is now misclassified.

(e) Explain the mechanism in three lines, referring to what squared error charges for.

U5.60 [$\star\star\star$][**NUM**] Complete the table and answer what follows.

p	odds	log-odds
0.05		
0.20		
0.50		
0.80		
0.95		

State the antisymmetry property you can see in the third column, and say what it implies about relabelling the two classes.

U5.61 [$\star\star\star$] Derive the logistic regression gradient from a blank page: the Bernoulli probability of one row, the likelihood, the log-likelihood, the three chain-rule factors, and the cancellation that produces $\partial\ell/\partial w_j = \sum_i (y_i - p_i)x_{ij}$. *Show the cancellation explicitly.*

U5.62 [$\star\star\star$][**NUM**] Take $X = \begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$, $y = (0, 0, 1, 1)$, $w = (0, 0)$, $\eta = 0.4$.

(a) Compute z, then p, then the log-likelihood. Express the per-row negative log-likelihood as a familiar constant.

(b) Compute $p - y$ and the gradient $X^\top(p - y)$.

(c) Perform one gradient step and give the new w.

(d) Compute the new z, the new p and the new log-likelihood. State the improvement.

(e) The intercept component of the gradient is exactly zero. Explain why.

U5.63 [$\star\star\star$][**NUM**] A logistic model has coefficient $\beta = \ln 3$ on a feature.

(a) State the odds ratio.

(b) For starting probabilities 0.1, 0.2, 0.5 and 0.8, compute the new probability after a one-unit increase, via the odds.

(c) Give Δp in each case and state where it is largest.

(d) Compute the divide-by-four bound and confirm your table respects it.

(e) Correct this sentence: "a one-unit increase raises the probability by 200%".

U5.64 [⋆⋆][**NUM**] Compute $\sigma(z)$ and $\sigma(z)(1-\sigma(z))$ for $z=-3,-1.5,0,1.5,3$. Verify $\sigma(-z)=1-\sigma(z)$ on your table, state where the derivative is largest and its value there, and explain what that maximum has to do with the divide-by-four rule.

U5.65 [⋆⋆⋆][**NUM**] A screening model gives these held-out counts.

threshold	TN	FP	FN	TP
0.10	90	20	1	39
0.25	98	12	3	37
0.50	105	5	7	33
0.75	109	1	12	28

(a) Compute accuracy, precision, recall and F_1 at every threshold.

(b) A missed cancer costs ten times a false alarm. Compute $10\,\mathrm{FN}+\mathrm{FP}$ for each row.

(c) State the threshold chosen by accuracy and the threshold chosen by cost, and say which you would deploy.

(d) The ROC–AUC is the same for all four rows. Explain why.

U5.66 [⋆⋆] A logistic fit returns coefficients $(48.2,-31.7,0.4)$, an accuracy of 1.0000 and a `ConvergenceWarning`.

(a) Name the phenomenon.

(b) Explain what happens to the log-likelihood as $\|w\|\to\infty$, and why the maximum likelihood estimate does not exist.

(c) State why accuracy cannot detect this.

(d) Name the fix, state what it costs, and say which lecture covers it.

U5.67 [⋆⋆] Explain why there is no closed form for logistic regression. Write both sets of equations side by side, say precisely what makes one solvable and the other not, and name the two iterative methods used instead.

§C — Ten-mark questions

U5.68 [⋆⋆⋆] Give a complete account of least squares: the model and its assumptions; why squared vertical deviations; the full derivation including the second-order condition; the matrix form; the four properties of the fitted line; $\mathrm{Var}(b_1)=\sigma^2/S_{xx}$ and its design implication; and what least squares will not tell you. Include a worked numerical fit.

U5.69 [⋆⋆⋆] Give a complete account of maximum likelihood for the linear model: the probability model; the likelihood and log-likelihood; the central equivalence with least squares and why σ^2 is irrelevant to it; the ML estimate of σ^2, its bias, and the unbiased alternative with the degrees-of-freedom reason; and what changes under a Laplace noise model.

U5.70 [⋆⋆⋆] Give a complete account of R^2 and model adequacy: the decomposition; the two definitions of R^2 and the identity chain in simple regression; why R^2 cannot decrease with an added predictor; adjusted R^2 and its limits; negative held-out R^2; and why a residual plot must accompany every reported R^2. Use Anscombe's quartet as the illustration.

U5.71 [⋆⋆⋆] Give a complete account of over-fitting and its detection: what it is; why training error cannot detect it; the degree curve with numbers; why it depends on n as well as the model; cross-validation as the detector; why one validation split is not enough; and the rule about what must be refitted inside every fold.

U5.72 [⋆⋆⋆] Derive logistic regression completely: why a straight line fails on 0/1 labels, with a worked numerical demonstration; odds, log-odds and the inversion to the sigmoid; the Bernoulli likelihood and log-likelihood; the gradient with the cancellation shown in full; why there is no closed form; concavity and what it does and does not guarantee; and one gradient step worked by hand.

U5.73 [⋆⋆⋆] Explain the interpretation and deployment of a logistic model: the odds-ratio derivation; why e^{β} is multiplicative in the odds and not additive in the probability, with a worked table; the divide-by-four rule; why raw coefficients cannot be compared across features; the decision threshold as a choice made outside the model; and choosing that threshold under asymmetric costs with a worked confusion matrix.

U5.74 [⋆⋆] Compare least squares and logistic regression across: what is being modelled; the loss; whether a closed form exists; the shape of the objective; how the parameters are found; how a coefficient is interpreted; and what pathology each suffers from. Support with numbers from a worked fit of each.

U5.75 [⋆⋆] "State the noise distribution, write the likelihood, take logarithms, maximise." Show that this single recipe produces least squares under Gaussian noise, least absolute deviations under Laplace noise, and cross-entropy under a Bernoulli model. Do the derivation in each case.

§D — Statements to judge (2 marks each)

U5.76 [⋆⋆⋆] "Least squares is chosen because it is differentiable."

U5.77 [⋆⋆] "The residuals of a least-squares fit always sum to zero."

U5.78 [⋆⋆⋆] "$\hat{\sigma}^2_{\mathrm{ML}}$ is biased because we divide by n instead of $n-2$."

U5.79 [⋆⋆⋆] "R^2 measures how good the model is."

U5.80 [⋆⋆] "A higher R^2 means a better model, so add the predictor."

U5.81 [⋆⋆⋆] "e^{β_j} is the increase in the probability."

U5.82 [⋆⋆] "Logistic regression is not a linear model, because the sigmoid is not linear."

U5.83 [⋆⋆] "The model reached 100% accuracy, so the fit is sound."

U5.84 [⋆] "`LogisticRegression()` in scikit-learn returns the maximum likelihood estimate."

Mock paper B (90 minutes, 35 marks, scaled to 20)

Sit this one properly

Closed book. No computer. 90 minutes on a timer. Mock paper A is Section D of Assignment A1; this is the second of three. Do not read the questions before you start.

Section A — attempt all (5 × 2 = 10)

1. Write the Huber loss with parameter δ and state the one property that makes it a compromise between MSE and MAE. **[2 marks]**

2. State the stability condition on the learning rate for a quadratic with largest curvature L, and say what happens at exactly $\eta = 2/L$. **[2 marks]**

3. Give the formula for the probability that a row survives listwise deletion, and evaluate it for $p = 0.05$ and $d = 20$. **[2 marks]**

4. Name the four scalers and state which one a single extreme value destroys, with the reason. **[2 marks]**

5. State the four properties of a least-squares fitted line, and say which one requires the model to have an intercept. **[2 marks]**

Section B — attempt any 3 of 4 (3 × 5 = 15)

6. Perform three steps of **RMSprop** on $f(w) = (w-4)^2$ from $w_0 = 0$ with $\eta = 0.5$, $\gamma = 0.9$, $\varepsilon = 10^{-8}$. Tabulate t, g_t, s_t, $\sqrt{s_t}$, the step and w_t. Show that the first step equals $\eta/\sqrt{1-\gamma}$, and state at which step s_t first falls and why Adagrad's cannot. **[5 marks]**

7. A column reads $20, 21, 19, 22, 20, 21, 23, 19, 20, 22, 120, 125$. Compute the mean, sample sd, median, MAD, Q_1, Q_3, IQR and the 1.5×IQR fences. Score the value 120 under all three outlier rules and state which fire. Name the effect that explains the disagreement. **[5 marks]**

8. Fit a least-squares line to $x = (0, 1, 2, 3, 4)$, $y = (1, 2, 4, 5, 8)$: give S_{xx}, S_{xy}, b_1, b_0, the fitted values and the residuals; verify $\sum e_i = 0$ and $\sum x_i e_i = 0$; compute SSE, SSR, SST and R^2, checking SSR a second way. **[5 marks]**

9. Take X with rows $(1, -1), (1, 0), (1, 1), (1, 2)$, $y = (0, 0, 1, 1)$, $w = (0, 0)$ and $\eta = 0.4$. Compute p, the log-likelihood, the gradient $X^\top(p - y)$, one gradient step, and the new log-likelihood. Explain why the intercept component of the gradient is exactly zero. **[5 marks]**

Section C — compulsory (10)

10. Trace the development from batch gradient descent to Adam as a chain of fixes.

(a) For each of batch GD, SGD, mini-batch, momentum, Adagrad, RMSprop and Adam, state the problem it was invented to fix and write its update rule. **[4 marks]**

(b) Explain why a single scalar learning rate can fail for two entirely distinct reasons, and state which of the two momentum addresses. **[2 marks]**

(c) Derive Adam's bias correction for a constant gradient g, and show that the first corrected step is exactly $\pm\eta$. **[2 marks]**

(d) "Adam is better than SGD." Give the strongest case for this claim and the strongest case against it, and state what adaptive optimisers actually buy. **[2 marks]**

Mock paper C (90 minutes, 35 marks, scaled to 20)

Section A — attempt all ($5 \times 2 = 10$)

1. State Mitchell's definition of learning and name T, P and E for a credit-scoring system. **[2 marks]**

2. Write the binary cross-entropy formula and state what it evaluates to for a single example given probability 0.5 to the correct class. **[2 marks]**

3. Write the momentum update rule and give the effective step size in terms of η and β. **[2 marks]**

4. State the factor by which mean imputation shrinks a column's standard deviation, and evaluate it when 25% of values are missing. **[2 marks]**

5. Define odds and log-odds, give the range of each, and state which of the two a linear expression $w^\top x + b$ can safely produce. **[2 marks]**

Section B — attempt any 3 of 4 ($3 \times 5 = 15$)

6. A model gives $\hat{y} = (3.5, 5.0, 4.5, 8.0, 5.0)$ against $y = (3, 6, 2, 9, 5)$. Compute MSE, MAE, RMSE and the Huber loss at $\delta = 1.5$, showing the branch each point takes. Compute the share of the total squared error and of the total absolute error carried by the worst point, and comment. **[5 marks]**

7. A nominal colour variable is ordinal-encoded blue = 0, green = 1, red = 2, with target group means 30, 70 and 15. Compute S_{xy}, S_{xx}, the least-squares slope and intercept, the three predictions, SSE, SST and R^2. State what one-hot encoding would achieve and explain the difference. **[5 marks]**

8. A fit on five rows gives SSE = 1.10 with $n = 5$ and $\mathrm{SS_{tot}} = 30$. Compute R^2, $\hat{\sigma}^2_{\mathrm{ML}}$, the unbiased s^2 and s. Give the bias factor and the percentage understatement, and explain *why* the divisor is $n - 2$ — restating the formula earns nothing. **[5 marks]**

9. A logistic model has coefficient $\beta = \ln 3$. State the odds ratio; compute the new probability after a one-unit increase from starting probabilities 0.1, 0.2, 0.5 and 0.8, via the odds; state where Δp is largest; give the divide-by-four bound; and correct the sentence "a one-unit increase raises the probability by 200%". **[5 marks]**

Section C — compulsory (10)

10. Logistic regression, from the failure of the straight line to a deployed threshold.

(a) Fit a least-squares line to $x = (1, 2, 3, 4, 5, 6)$ mm with labels $y = (0, 0, 0, 1, 1, 1)$. Give the slope, the intercept, the six predictions and the 0.5 crossing. State two distinct things wrong with using this as a classifier. **[3 marks]**

(b) Derive the sigmoid by inverting the logit link, and state three properties of σ including its derivative. **[2 marks]**

(c) Derive $\partial\ell/\partial w_j = \sum_i (y_i - p_i)x_{ij}$ from the Bernoulli log-likelihood, showing the cancellation explicitly. **[3 marks]**

(d) There is no closed form for w. Say precisely why, contrasting the score equations with the normal equations, and name what concavity does and does not guarantee. **[2 marks]**

Eleven ways to lose marks you have earned

1. **Restating the definition instead of answering.** "The z-score rule detects outliers more than three standard deviations from the mean" is the definition. If the question asks why it failed, that sentence earns nothing.

2. **A number with no working.** Full marks need the steps. A correct final value alone typically earns 1 of 5.

3. **Working with no number.** The converse also fails. Finish the arithmetic.

4. **Naming without justifying.** "Use the IQR rule" is half an answer. *Why* is the other half, and it usually carries the mark.

5. **Forgetting the units.** An MAE of 1.0 in lakhs of rupees is not an MAE of 1.0.

6. **Answering a different question.** If asked for the effect on *precision*, do not write about accuracy.

7. **One example given twice.** Asked for two examples of leakage, "scaling on all the data" and "normalising before the split" are one.

8. **Ignoring the mark allocation.** A 2-mark question wants four lines. A 10-mark question wants structure, a worked illustration and a conclusion. Writing four lines for ten marks is the most common way to lose a grade.

9. **Skipping the numericals in revision.** They are the most predictable marks on the paper and the ones most often left blank.

10. **Stopping a derivation at the first-order conditions.** Both the least-squares derivation and the maximum-likelihood one carry marks for the *second-order* check. Setting a derivative to zero finds a stationary point; it does not establish that you have found a minimum.

11. **Saying "odds" when you mean "probability", or the reverse.** e^{β} multiplies the *odds*. It is not an increase in probability, and no amount of surrounding correct working rescues a sentence that says it is.

Answer key released after AS1. Until then, the lecture notes for **L1–L10** carry their own practice questions *with* full solutions — work those first if you want something to check yourself against.